

French Motor Insurance: Claim Frequency, Severity, and Expected Cost Modeling

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Abstract

We estimate annual expected claim cost for French motor third-party liability policies by modeling claim frequency and mean claim severity separately. We reconciled 677991 policy records with 26444 claim records, examined frequency and severity patterns, compared candidate models using five policy-level development folds, and evaluated the selected specifications on reserved policies. Histogram gradient-boosted decision trees fitted with Poisson loss and a Gamma log-link GLM predicted 11.663 million in test loss against 11.734 million observed, an observed-to-predicted ratio of 1.006. Frequency performance was consistent between development and test data. Severity was less reliably calibrated across risk groups and remained sensitive to rare large claims. Comparisons with more flexible severity models and an explicit body/tail model informed the choice of a comparatively stable conditional-mean estimator.

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1 Introduction

An insurer needs to estimate both how often a policy will generate claims and how much those claims will cost. For policy characteristics X , let $\lambda(X)$ denote expected claims per exposure-year and let $\mu(X)$ denote expected amount per claim. The annual expected claim cost, or pure premium, is

$$PP(X) = \lambda(X)\mu(X).$$

For a policy observed for e years, expected loss over that period is $ePP(X)$. Pure premium represents expected claim cost and does not include expenses, profit provisions, or other components of a charged premium.

The selected frequency model uses histogram gradient-boosted decision trees fitted with Poisson loss; the severity model is a Gamma GLM with a log link. Their combined prediction is within 0.6% of observed aggregate loss on the reserved policies. The model choice reflects a difference in what the data support. Frequency patterns are relatively stable across validation folds, whereas a few large claims have substantial influence on estimates of mean severity. A separate Gamma-body/GPD-tail model made that uncertainty more explicit but did not improve the stability of expected-amount prediction. Close portfolio-level calibration is therefore useful evidence for the pricing model, while the weaker severity calibration across risk groups remains an important qualification.

The accompanying Python notebooks contain the data checks, exploratory summaries, fitted specifications, and diagnostics underlying this report. Notebooks 01–03 cover data integrity and exploratory frequency and severity analysis. Notebook 04 develops the initial GLMs and establishes the policy-level split. Notebook 05 compares predictive models within the development sample, and Notebook 06 investigates explicit body/tail severity models. Notebook 07 refits the selected models and evaluates their combined expected-loss predictions on the reserved policies.

2 Data Sources and Integrity Checks

The analysis uses the `freMTPL2freq` and `freMTPL2sev` datasets from the `CASdatasets` R package. Both were exported to CSV with `write.csv(..., row.names = FALSE)` and examined in Python with `pandas`. The frequency file has 677991 rows and 12 columns. Each row represents a policy exposure period and includes a policy identifier, claim count, exposure in years, and driver, vehicle, and geographic attributes. The severity file has 26444 rows and two columns: the policy identifier and a claim amount. It can contain multiple claim records for one policy.

Before analyzing or combining the files, we checked their schemas, missing values, policy identifiers, exposure and outcome ranges, and the agreement between policy claim counts and claim records. Neither file has missing values. All 677991 policy identifiers in the frequency file are unique. The severity file contains 24944 distinct policy identifiers, all of which appear in the frequency file. Repeated identifiers in the severity file are expected when a policy has multiple claims.

One policy identifier in the frequency CSV is written in scientific notation, causing `pandas` to infer a floating-point type for the identifier column unless a type is specified. We read `IDpol` as

text in both files for the linkage checks. The affected identifier represents exactly 100000, and no severity records were unmatched.

All policy exposures are positive, with a median of 0.49 years. There are 1224 policies with exposure above one year, and the maximum is 2.01 years. All claim counts are nonnegative and all recorded claim amounts are positive. Claim amounts are highly right-skewed: the median is 1172, while the maximum exceeds 4.07 million. These observations call for appropriate exposure adjustment and attention to large claims in subsequent analysis, but do not alone justify dropping records.

Finally, we counted severity records by policy and compared that count with the frequency file's `ClaimNb`. The frequency file reports 26444 claims in total, exactly matching the number of severity rows, and the counts agree for every individual policy. This comparison established that the files can be linked consistently; all records were retained for the subsequent analyses. The exploratory summaries in the next two sections used the full supplied portfolio before the policy split was established. Later, policies were assigned to an 80% development sample and a reserved 20% test sample, with all claims on a policy following its assignment. The reserved policies were excluded from model fitting and candidate selection, but were not wholly unseen during exploratory analysis. Final test metrics are consequently out-of-sample assessments for fitting and selection, subject to that qualification.

3 Exploratory Analysis of Claim Frequency

The portfolio contains 26444 claims over 358482.84 exposure-years, giving an observed frequency of 7.38 claims per 100 exposure-years. For each group below, frequency is calculated as the group's total claims divided by its total exposure. The dashed lines in the figures mark the portfolio rate. These are descriptive comparisons of one characteristic at a time; they do not adjust for differences in other policy characteristics.

Most policies reported no claims during their observed exposure period. Of the 677991 policy records, 24944 (3.68%) reported at least one claim. This percentage is not an annual claim frequency: exposure lengths vary, and some policies have multiple claims.

3.1 Driver Characteristics

Drivers aged 18–24 have an observed rate of 16.10 claims per 100 exposure-years, more than twice the portfolio rate. This group accounts for 2011 claims over 12487 exposure-years. As Figure 1 shows, the other broad age groups have substantially lower rates, ranging from 5.81 to 7.70.

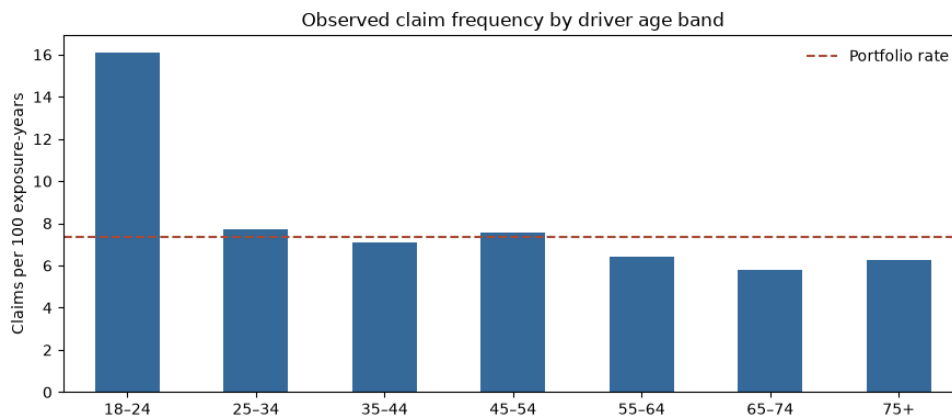


Figure 1: Observed claim frequency by driver age band. The dashed line indicates the portfolio rate of 7.38 claims per 100 exposure-years.

The broad grouping masks a continuing decline throughout the twenties rather than a sudden change at age 25. In Figure 2, the rate falls from 30.37 at age 18 to 11.51 at age 24 and 7.92 at age 27. Differences among later ages are comparatively modest. Individual ages with fewer than 100 exposure-years were omitted from the plot because their observed rates are more variable.

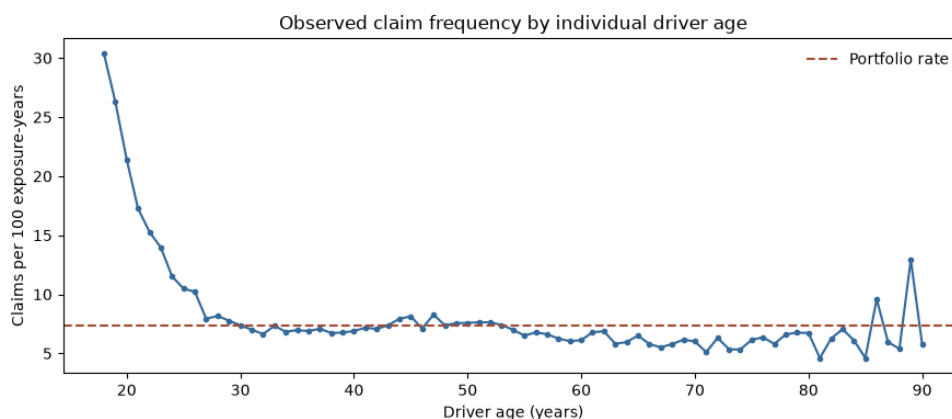


Figure 2: Observed claim frequency by individual driver age. Ages with fewer than 100 exposure-years are omitted from the plot.

Bonus-malus has an even stronger descriptive association with claim frequency. In the broad grouping, the rate rises from 6.23 for values below 75 to 45.60 for values of 125 or greater. Figure 3 shows that rates remain around 10–12 across much of the 60–99 range, then rise substantially above 100. The highest values have much less exposure than the lower groups, so their observed rates warrant more caution. Bonus-malus also reflects prior claims experience, distinguishing its interpretation from that of an ordinary demographic characteristic.

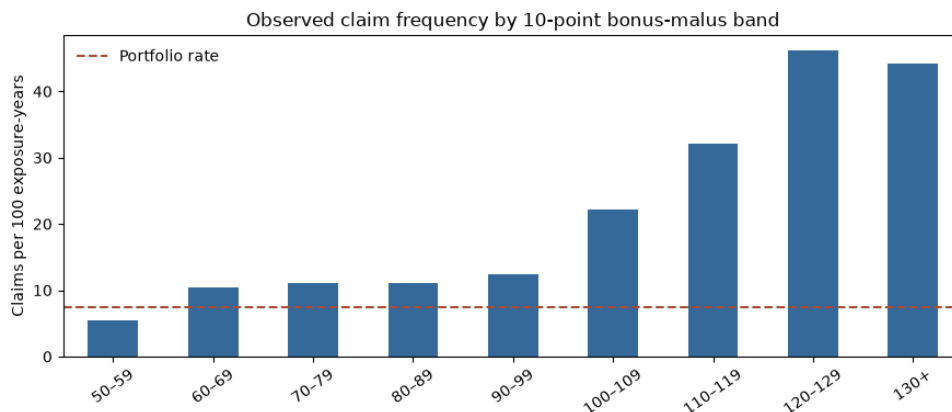


Figure 3: Observed claim frequency by ten-point bonus-malus bands, with values of 130 or greater grouped together.

3.2 Vehicle Characteristics

Vehicle characteristics show smaller and less consistent differences. Diesel policies have 7.88 claims per 100 exposure-years, compared with 6.92 for regular-fuel policies. Figure 4 shows broadly similar rates for vehicles aged 0 through 20 years, followed by a lower rate of 4.02 for vehicles aged 21 years or more. The latter group contributes 5147 exposure-years, considerably less than the other age groups. Vehicle power does not show a clear monotonic pattern. Frequencies also vary among the coded vehicle brands, although the codes alone offer limited substantive interpretation.

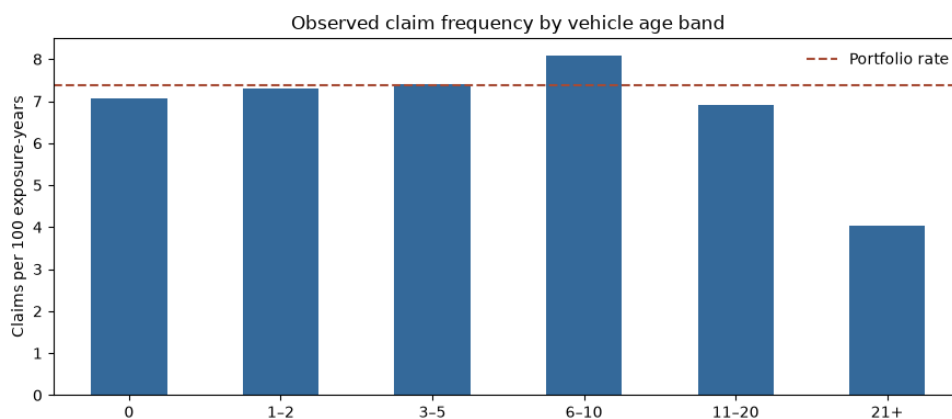


Figure 4: Observed claim frequency by vehicle age band.

3.3 Geographic Characteristics

The geographic comparisons show a clearer pattern. Area is ordered from rural category A to urban-centre category F. As Figure 5 shows, observed frequency rises from 5.43 in A to 9.59 in E, with a similar rate of 9.52 in F.

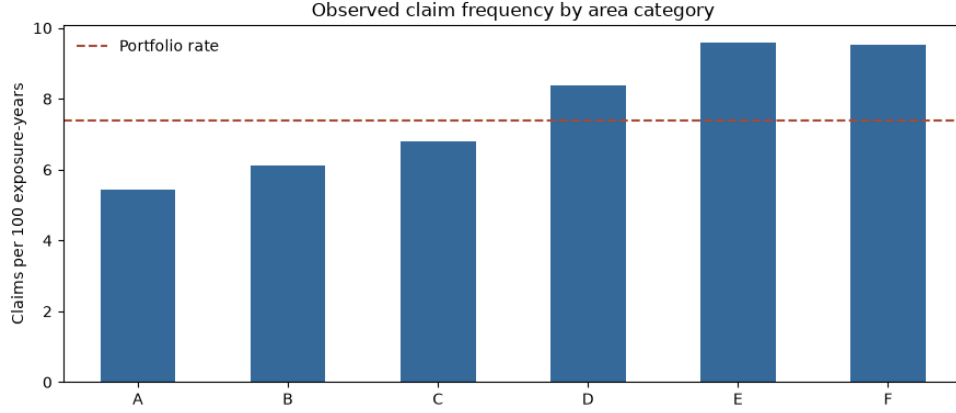


Figure 5: Observed claim frequency across the ordered rural-to-urban Area categories.

Population density exhibits a similar association. When policies are divided into five approximately equal-sized groups, observed frequency rises from 5.55 in the lowest-density group to 9.61 in the highest. Figure 6 uses finer, roughly logarithmic density ranges. It suggests that the increase becomes less pronounced at the highest densities. The group with density of 1–2 inhabitants per square kilometer has fewer than 100 exposure-years and is omitted from the plot.

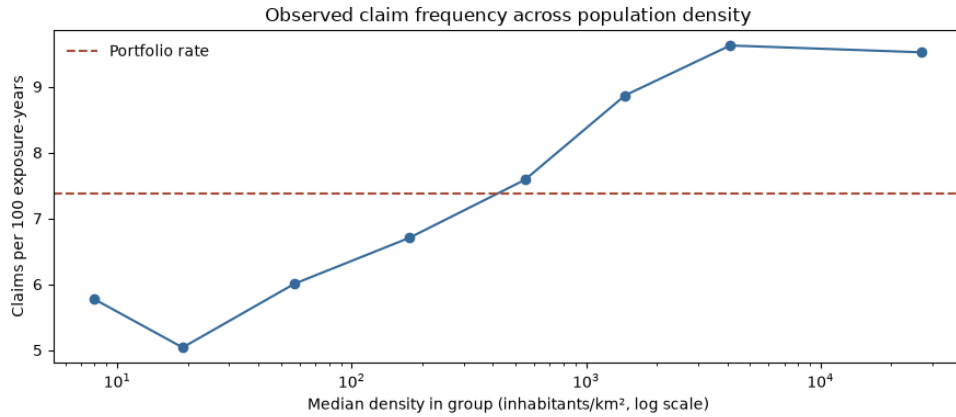


Figure 6: Observed claim frequency across population-density groups. Each group is plotted at its median density on a logarithmic horizontal axis.

Area and density capture overlapping aspects of geography: median density rises from 27 inhabitants per square kilometer in Area A to 27000 in Area F. Mean exposure per policy also declines from 0.60 years in A to 0.45 in F, reinforcing the need to compare claims per exposure-year. Regional frequencies vary as well, but differences in urbanization and other portfolio characteristics may contribute to those comparisons. The marginal patterns in this section identify relationships to examine jointly in a multivariable frequency model; they should not be interpreted as adjusted effects.

4 Exploratory Analysis of Claim Severity

Claim severity is the amount of a recorded claim, conditional on a claim occurring. We joined the 26444 claim records to their policy characteristics using IDpol. Every claim matched exactly one policy record. A policy with multiple claims contributes multiple observations to this analysis. The group means below are therefore averages over claims, rather than over policies. As in the frequency analysis, the comparisons consider one characteristic at a time and are not adjusted for other characteristics.

4.1 Distribution and Concentration of Claim Amounts

The recorded claims total 59909216.50, giving a mean amount of 2265.51 per claim. The median is 1172.00, considerably below the mean. Figure 7 displays the long right tail using logarithmic axes. The 99th percentile is 16451.22, while the largest claim is 4075400.56. There is also substantial concentration around particular recorded amounts: 1204.00, 1128.12, and 1172.00 each occur thousands of times. The dataset documentation notes that some claim amounts are fixed under French settlement conventions, which may help explain these repeated values.

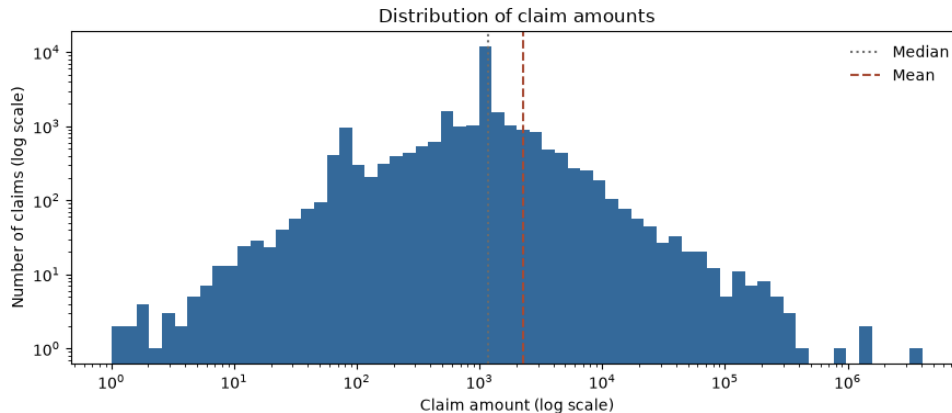


Figure 7: Distribution of recorded claim amounts, with logarithmic axes. The vertical lines indicate the median and mean.

Large claims account for a disproportionate share of total cost. The largest 27 claims, approximately 0.1% of all claims, contribute 21.6% of the total amount. The largest 1% contribute 38.0%, and the largest 5% contribute 52.1%. Figure 8 shows the same concentration cumulatively, ordering claims from smallest to largest. This concentration makes the mean, which will enter the frequency–severity calculation of expected claim cost, sensitive to a relatively small number of observations.

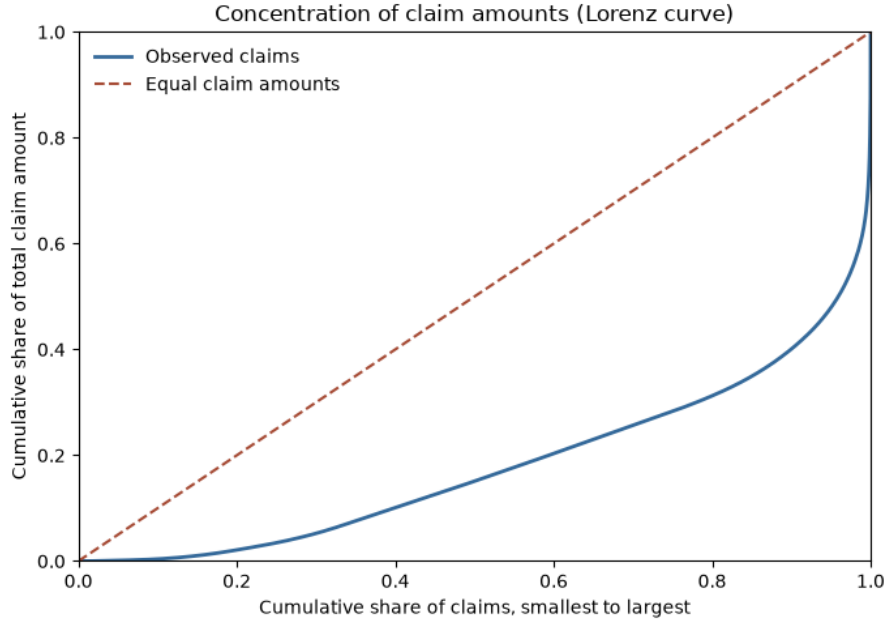


Figure 8: Cumulative share of total claim amount against the cumulative share of claims, ordered from smallest to largest. The dashed diagonal represents equal claim amounts.

4.2 Severity by Policy Characteristics

Figure 9 compares mean claim amounts across driver age groups. Claims associated with drivers aged 18–24 have the highest mean, at 5838.56, compared with the portfolio mean of 2265.51. Yet the median for this group is 1172.00, the same as or close to the medians of the other groups. The unusually high mean thus reflects the upper tail more than a comparable shift in typical claim amounts.

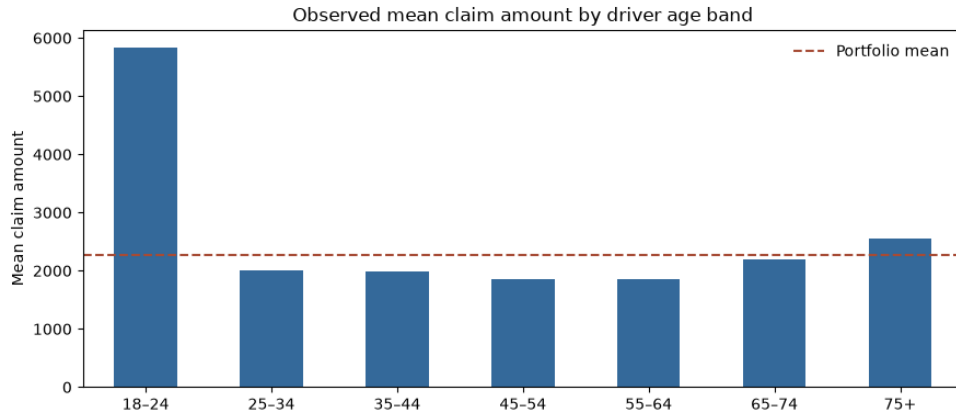


Figure 9: Mean claim amount by driver age band. The dashed line indicates the portfolio mean of 2265.51.

Bonus-malus shows a different pattern from the steady increase in claim frequency observed earlier. The mean amount is 5536.74 for claims in the 100–124 group, but falls to 2183.49 in the 125+ group, as shown in Figure 10. The median is 1172.00 in every broad bonus-malus group.

The high mean for the 100–124 group should therefore not be interpreted as evidence of a similarly large difference throughout its claim-amount distribution.

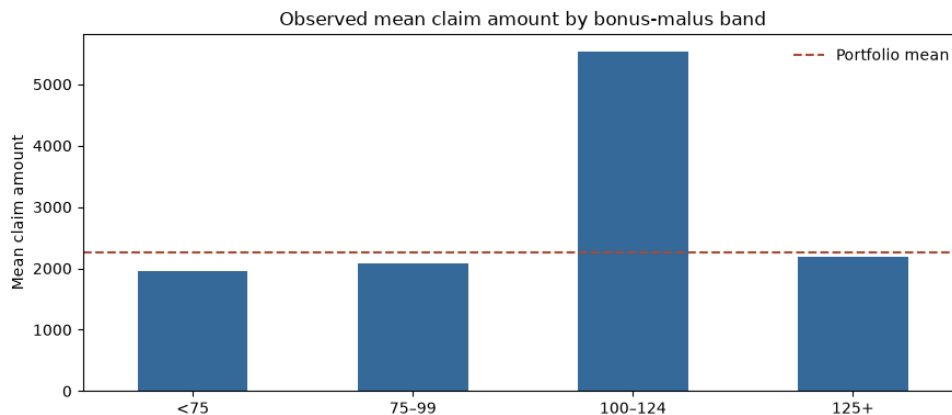


Figure 10: Mean claim amount by bonus-malus band.

Mean amounts are relatively similar across most vehicle age groups, although the mean for vehicles aged 11–20 years reaches 2917.02 (Figure 11). Their median claim amount is 1144.64, slightly below the portfolio median. The 21+ group contains only 207 claims, so its mean is particularly susceptible to individual large observations. Regular-fuel policies have a mean claim amount of 2486.88, compared with 2051.65 for diesel policies. Vehicle power and coded brand also show variation, but their marginal means do not reveal a simple pattern.

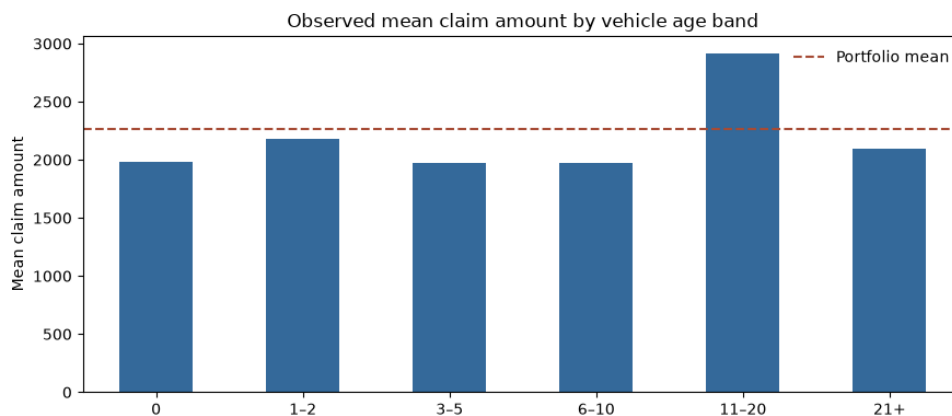


Figure 11: Mean claim amount by vehicle age band.

Unlike claim frequency, mean severity does not rise consistently from rural to urban areas. Area B has the highest mean, at 3370.29, while Area F has the lowest, at 1524.04 (Figure 12). The median is 1172.00 in every Area category. The density quintiles likewise show no steady progression in mean amount, as Figure 13 illustrates. These quintiles use the same policy-level density cut points as the frequency analysis, permitting direct comparison of the group definitions. Regional means vary as well, but estimates for regions with few claims may be dominated by isolated large losses.

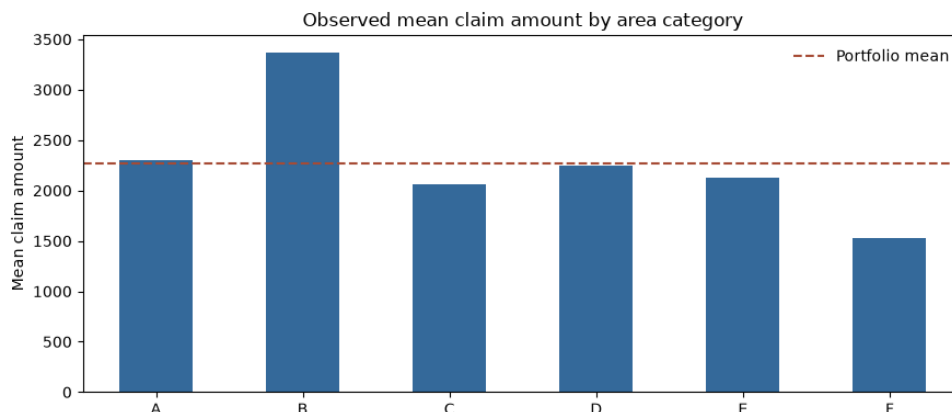


Figure 12: Mean claim amount by rural-to-urban Area category.

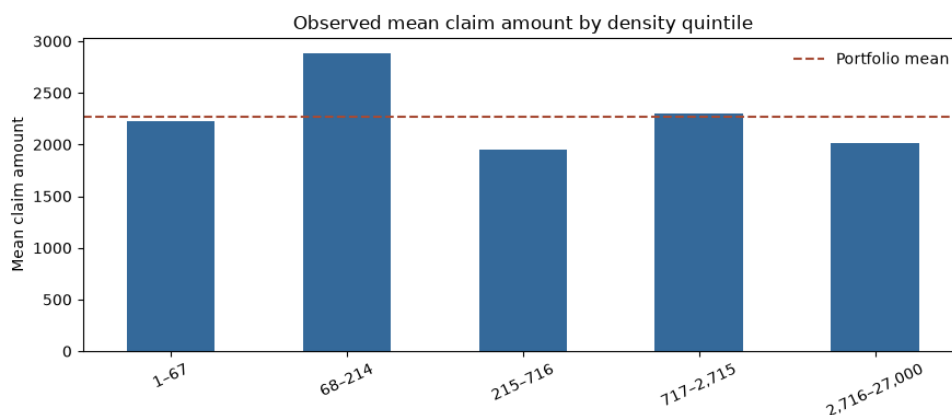


Figure 13: Mean claim amount by population-density quintile. The cut points are based on policy records and match those used in the frequency analysis.

4.3 Influence of Large Claims

To assess how much the upper tail affects these comparisons, we performed a sensitivity calculation in which each amount above the portfolio-wide 99th percentile of 16451.22 was set to that value. This changes 265 claims. The resulting portfolio mean is 1569.16, a reduction of 30.7% from the original mean.

Several prominent group differences also become much smaller under this calculation. The mean for drivers aged 18–24 falls from 5838.56 to 1772.92; for bonus-malus values of 100–124, from 5536.74 to 1743.97; and for vehicles aged 11–20 years, from 2917.02 to 1480.77. Area B falls from 3370.29 to 1565.89. These changes show that a relatively small number of large claims has a substantial effect on the observed subgroup means.

The cap is used only to diagnose sensitivity. Large claims remain part of the observed cost and the severity modeling target. The model comparisons below assess predictions on the original claim-amount scale and examine whether apparent group differences remain useful when policy characteristics are considered jointly.

5 Initial Frequency and Severity Models

As a first modeling step, we fitted a Poisson GLM for policy claim counts with log exposure as an offset and a Gamma log-link GLM for positive claim amounts. Both models included driver, vehicle, and geographic characteristics, with spline terms for driver age and bonus-malus. Policies were divided into an 80% development sample and a reserved 20% test sample; all claims from a policy followed that policy’s assignment. Five policy-level folds were then established within the development sample for subsequent model comparison.

Figure 14 illustrates the initial adjusted age and bonus-malus responses for a reference policy. These curves prompted a closer examination of stability across fitting folds. In one severity fold, the fitting claims reached a bonus-malus value of only 190, while a validation claim had a value of 228. The unrestricted spline predicted an amount exceeding two billion for that otherwise ordinary claim. This was extrapolation of the fitted curve beyond its claim-level support, amplified by the log link. The initial GLMs therefore served as benchmarks and diagnostics rather than final model choices; the later comparison used simpler conventional predictor forms and examined fold-level behavior explicitly.

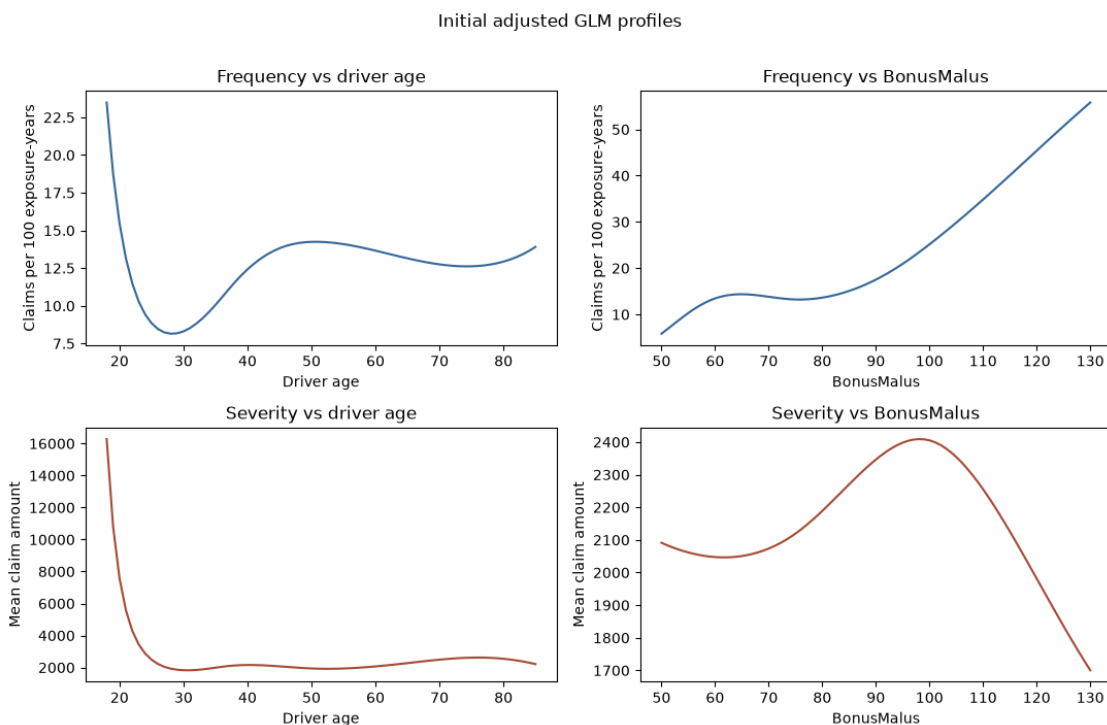


Figure 14: Initial adjusted frequency and severity responses to driver age and bonus-malus for an illustrative reference policy. These exploratory spline models were not selected for final evaluation.

6 Predictive Model Comparison

The exploratory analyses suggest that claim frequency and claim severity present different modeling problems. This section compares candidate models using five-fold cross-validation within an 80% development sample. All claims associated with a policy remain in the same fold, and each validation prediction is made by a model fitted without that fold. The remaining 20% of policies

are reserved for final evaluation and are not used here. The development sample contains 542830 policies and 21079 claims.

For frequency, we compared Poisson and Negative Binomial generalized linear models (GLMs) with histogram gradient-boosted decision trees fitted using a Poisson loss. The main change from the GLMs is the use of trees to learn nonlinear relationships and interactions. Policy exposure enters the GLMs as an offset and the boosted model as a weight on annualized claim frequency. For severity, we compared a Gamma GLM, boosted decision trees fitted with Gamma loss, and a regression on log claim amount with two methods of converting predictions back to expected amounts. The GLMs use transformed driver age, bonus-malus, vehicle age, and density, together with vehicle power, fuel type, and region. The boosted models use the corresponding numerical variables without those transformations. Their settings were fixed for this comparison rather than tuned to the validation results.

6.1 Claim Frequency

Table 1 summarizes the out-of-fold frequency results. The Poisson and Negative Binomial GLMs have nearly identical mean Poisson deviance. Boosted Poisson reduces deviance from approximately 0.24305 to 0.23716, an improvement of 2.4%. This improvement occurs in every validation fold, as shown in Figure 15. All three models predict approximately the correct aggregate number of claims.

Table 1: Out-of-fold frequency performance on the development sample. O/P is the ratio of total observed to total predicted claims.

Model	Mean Poisson Deviance	Predicted Claims	O/P
Poisson GLM	0.24305	21080	1.000
Negative Binomial GLM	0.24305	21174	0.996
Boosted Poisson	0.23716	21064	1.001

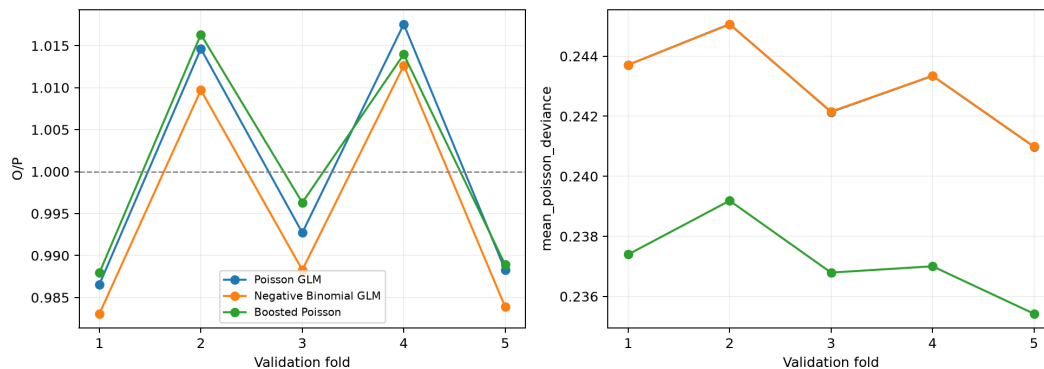


Figure 15: Observed-to-predicted claim ratios and mean Poisson deviance across the five validation folds.

Aggregate agreement alone does not show whether a model distinguishes lower- and higher-frequency policies. Figure 16 groups policies into deciles of each model’s predicted annual frequency. Observed and predicted rates are calculated using total claims and total exposure within each decile. The boosted model has observed-to-predicted ratios of approximately 0.95–1.06 across its deciles,

compared with approximately 0.91–1.12 for the Poisson GLM. It still underpredicts claims in its highest-risk decile by about 6%, but its calibration across the groups is generally closer.

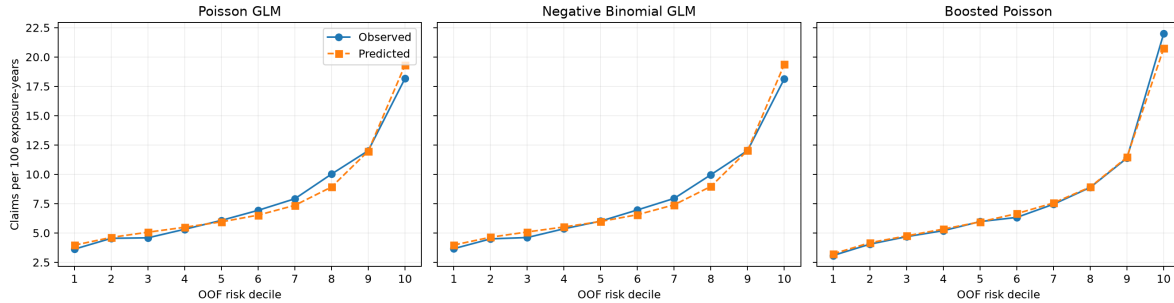


Figure 16: Out-of-fold calibration across deciles of predicted annual claim frequency. Each model defines its own deciles.

The fitted response curves help explain the difference between the models. The frequency GLMs impose relatively simple, monotone effects through their chosen transformations. Boosting allows a different age pattern, broadly consistent across the five fitting folds, as shown in Figure 17. Its responses to bonus-malus and density are shown in Figures 18 and 19. The density curves are particularly consistent across folds, while the sparsely populated extremes of age and bonus-malus vary more. These curves describe fitted model behavior and should not be interpreted as causal effects.

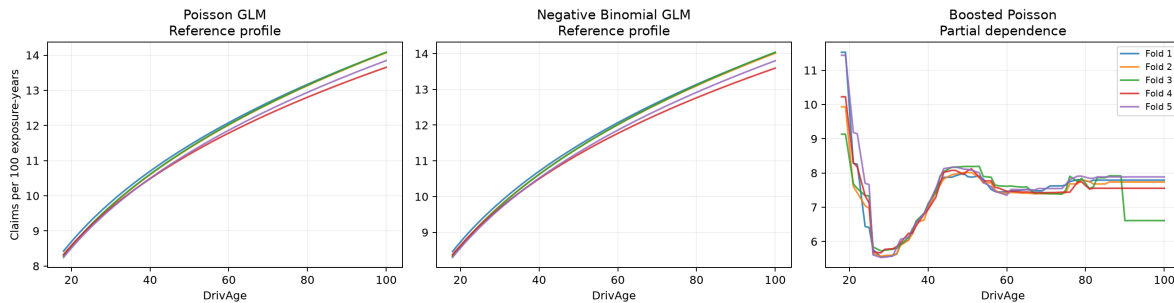


Figure 17: Fitted frequency responses to driver age across five fitting folds. The GLM panels use a reference policy profile; the boosted-model panel averages predictions over a sample of background policies.

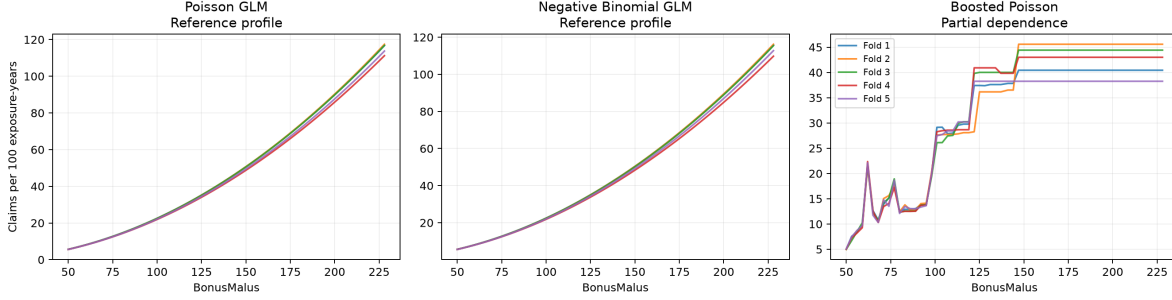


Figure 18: Fitted frequency responses to bonus-malus across five fitting folds, using the same model-specific response conventions as Figure 17.

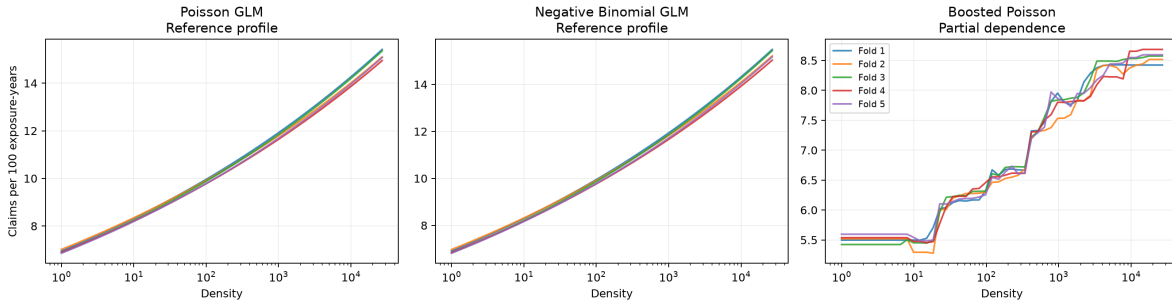


Figure 19: Fitted frequency responses to population density across five fitting folds.

The Negative Binomial specification does not improve out-of-fold prediction of the conditional mean relative to the Poisson GLM, despite evidence of overdispersion in the policy claim counts. Boosted Poisson has lower deviance in every fold and better calibration across predicted-risk groups. We therefore selected it as the development-stage frequency model.

6.2 Claim Severity

Severity model comparison requires particular care because the intended prediction is the arithmetic mean claim amount, while a small number of large claims accounts for a substantial share of total loss. Table 2 reports out-of-fold predictions on the original amount scale. The Gamma GLM predicts a mean of 2227.68 against an observed mean of 2285.46. Boosted Gamma has a lower mean absolute error (MAE), but predicts a mean of only 1798.72 and consequently understates aggregate loss by about 21%. Root mean squared error (RMSE) is approximately 32040 for every candidate, reflecting its strong sensitivity to the largest claims.

Table 2: Out-of-fold severity performance on the development claims. O/P is the ratio of total observed to total predicted claim amount.

Model	Predicted Mean	O/P	MAE	RMSE
Gamma GLM	2228	1.026	2289	32039
Boosted Gamma	1799	1.271	1981	32043
Lognormal, normal correction	1778	1.286	1946	32050
Lognormal, smearing correction	2319	0.985	2333	32048

The two lognormal rows come from the same regression of log claim amount. Exponentiating its fitted value alone would estimate a conditional median under the usual lognormal assumptions, so a correction is required to target the arithmetic mean. The normal-theory correction multiplies by $\exp(\hat{\sigma}^2/2)$; empirical smearing instead multiplies by the average of $\exp(e_i)$ over fitting residuals. Both correction factors are estimated within each fitting fold, without using its validation claims.

This choice has a large effect. The normal-theory factors range from 1.855 to 1.873, while the smearing factors range from 2.264 to 2.557. Normal-theory retransformation produces the lowest MAE but predicts an aggregate amount substantially below that observed. Smearing brings the pooled predicted mean close to the observed mean, but one extreme fitting residual contributes between 4.2% and 11.7% of the smearing sum, depending on the fold. A relatively stable regression on log amounts can therefore yield quite different estimates of expected claim amount depending on how it is retransformed.

Figure 20 compares observed and predicted amounts across predicted-severity quintiles. None of the candidates produces a convincing progression of observed mean claim amounts through these groups. In particular, the close aggregate agreement of the smearing version does not imply that it identifies which claims will produce the largest losses. Individual catastrophic claims materially affect some quintile means.

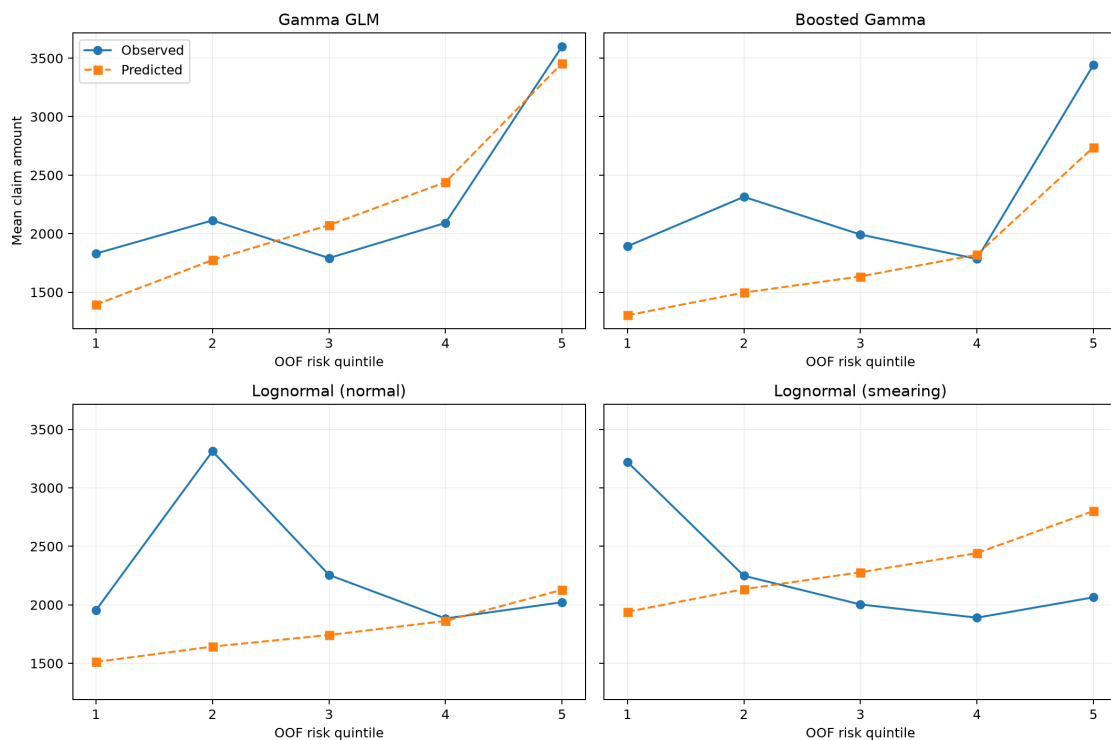


Figure 20: Out-of-fold calibration across quintiles of predicted mean claim amount. Each candidate defines its own quintiles.

Figure 21 shows substantial variation in observed mean severity across validation folds. This variation is much greater than that seen in the predicted fold means, and the direction of aggregate calibration changes from fold to fold. The reported fold variation describes these five particular splits; because their fitting samples overlap, it should not be treated as a standard error.

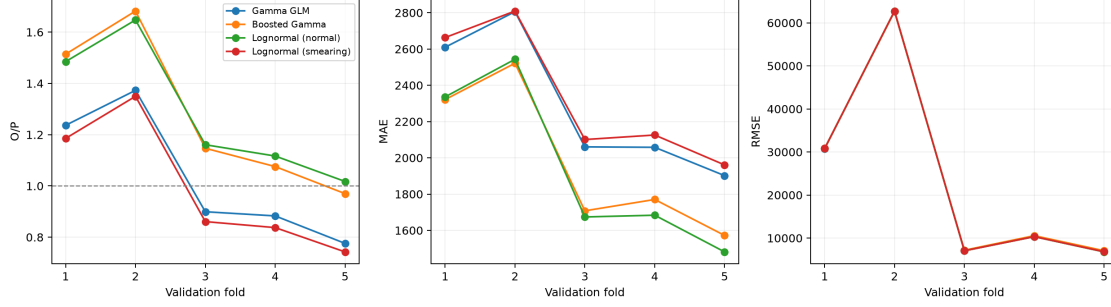


Figure 21: Observed and predicted severity summaries across the five validation folds.

To measure the influence of extreme observations, we repeated the evaluation after excluding the largest observed one, five, or 20 claims separately within each validation fold, leaving the predictions fixed. Excluding just one claim per fold removes five claims overall but 13.0% of observed loss. Excluding five per fold removes 23.2%, while excluding 20 per fold removes 32.7%. Figure 22 shows how strongly the resulting error and calibration measures change. For example, the Gamma GLM's pooled O/P falls from 1.026 with all claims to 0.893 after removing one per fold and 0.694 after removing 20 per fold.

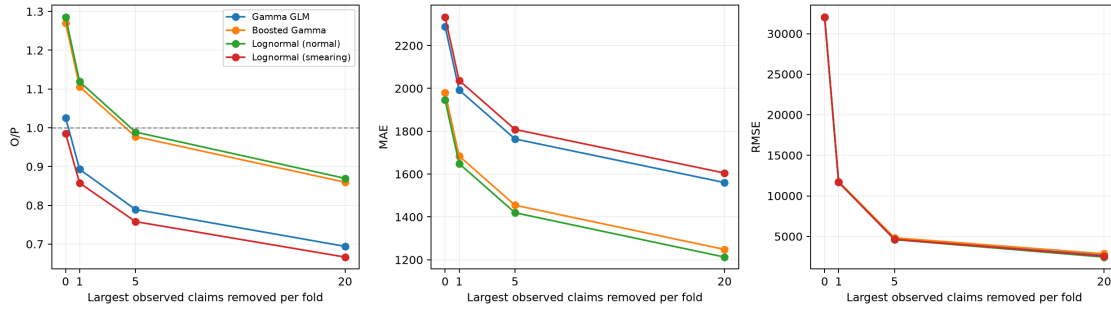


Figure 22: Sensitivity of out-of-fold severity performance to diagnostic exclusion of the largest observed claims within each validation fold. Models and predictions are held fixed.

These exclusions are diagnostic only. The large claims are genuine observed losses and remain part of the intended severity target. The exercise shows why a model's advantage on ordinary claims, or its aggregate agreement in this particular set of folds, does not by itself establish reliable prediction of expected loss.

Finally, Figures 23 and 24 compare fitted severity responses across folds. Both the Gamma GLM and boosted Gamma model show appreciable variation in the young-driver range. The lognormal response shapes are steadier under normal-theory retransformation, but their level changes more under the tail-sensitive smearing correction. Stability of a fitted curve cannot resolve the calibration problem on the original amount scale.

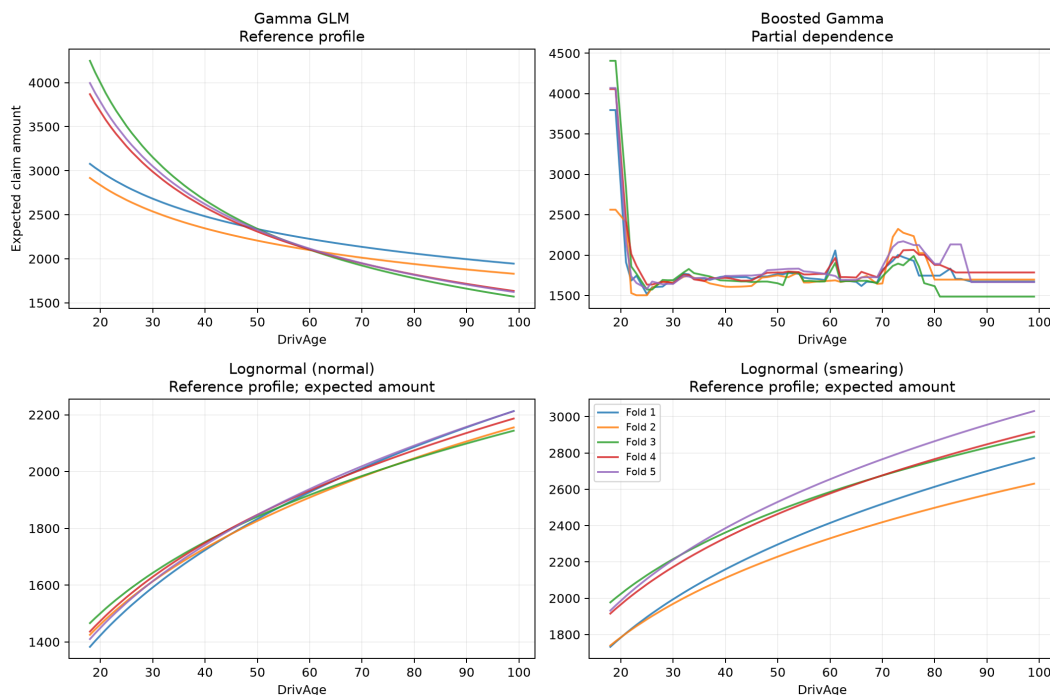


Figure 23: Fitted severity responses to driver age across five fitting folds. GLM and lognormal panels use a reference policy profile; the boosted-model panel averages over background claims.

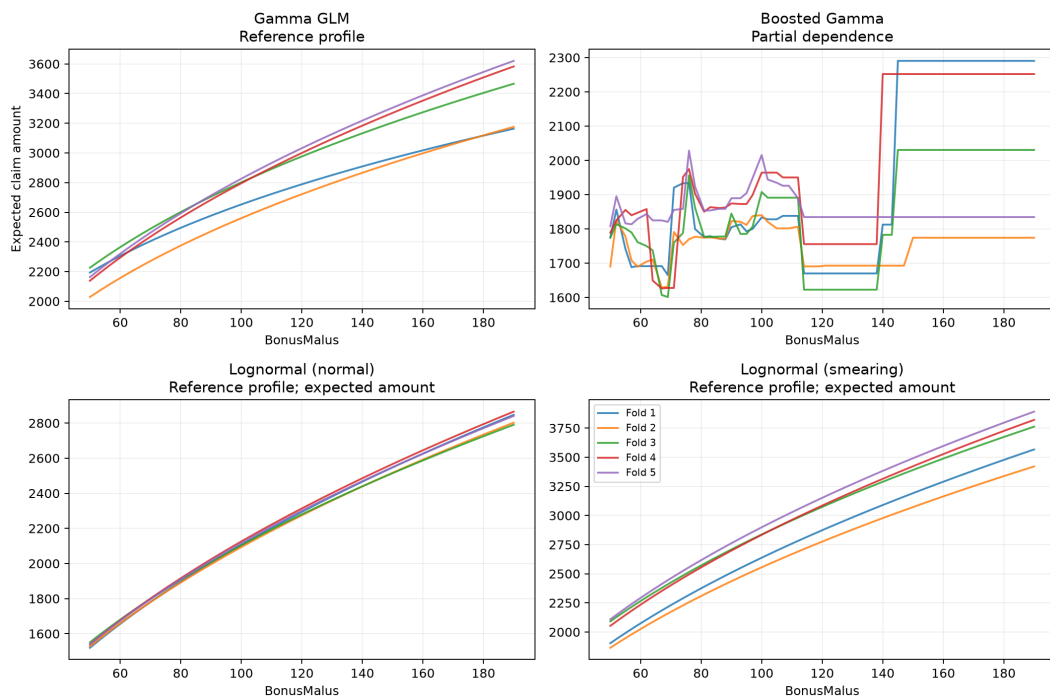


Figure 24: Fitted severity responses to bonus-malus across five fitting folds, using the same model-specific response conventions as Figure 23.

The Gamma GLM emerged as the strongest conventional severity candidate for expected-cost estimation, although its fold and tail diagnostics did not establish that its fitted distribution adequately describes extreme claims. We therefore examined an explicit large-loss model before fixing the severity specification.

7 Modeling Extreme Claim Severity

The preceding comparison showed that a few very large claims have a substantial effect on estimated mean severity. We therefore tested whether separating ordinary claims from extreme losses would improve predictions of expected claim amount. This experiment used the same five policy-level folds within the development sample. The reserved test sample was not used in this experiment.

For a threshold u , the model has three components: a Gamma distribution fitted to claims at or below u , a logistic model for the probability $p(X)$ that a claim exceeds u , and a generalized Pareto distribution (GPD) for the amount exceeding u . The Gamma body likelihood accounts for truncation at the threshold. If its conditional mean is denoted by $\mu_B(X)$ and the GPD has scale β and shape ξ , the combined expected severity is

$$\mu(X) = (1 - p(X))\mu_B(X) + p(X) \left(u + \frac{\beta}{1 - \xi} \right),$$

provided $\xi < 1$. The GPD scale and shape are constant across policies in this experiment; policy characteristics can affect the body mean and the probability of exceeding the threshold.

We compared thresholds of 5000, 10000, and 20000, fixed before fitting. At each threshold, a small sequence of models adds vehicle power, then log driver age and log bonus-malus, to the body and exceedance-probability components. Raising the threshold leaves fewer claims from which to estimate the GPD: each fitting fold has roughly 745–809 exceedances above 5000, 290–316 above 10000, and only 133–146 above 20000. Figure 25 displays mean excess over a range of diagnostic thresholds, while Figure 26 compares fitted tail parameters across the three candidate thresholds.

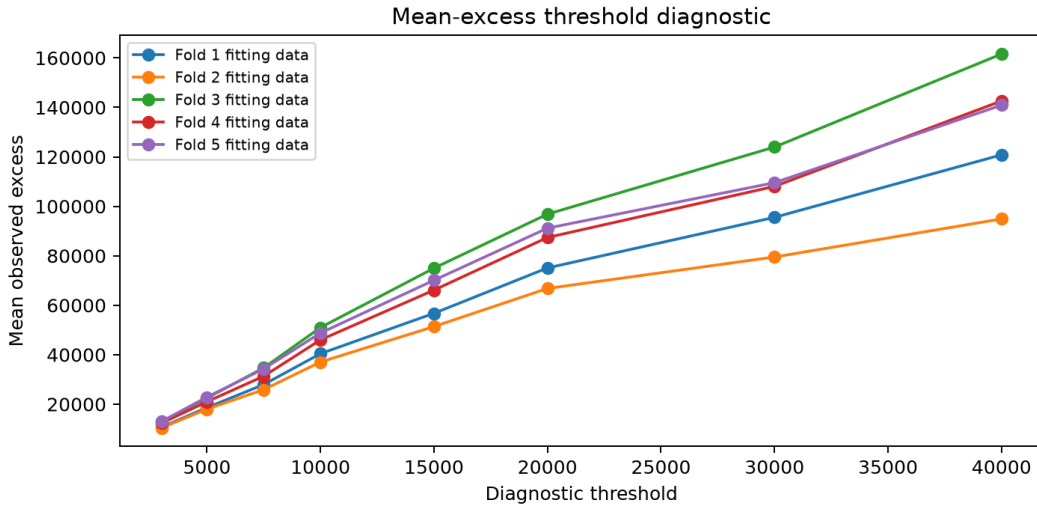


Figure 25: Mean excess above diagnostic thresholds across the five fitting samples.

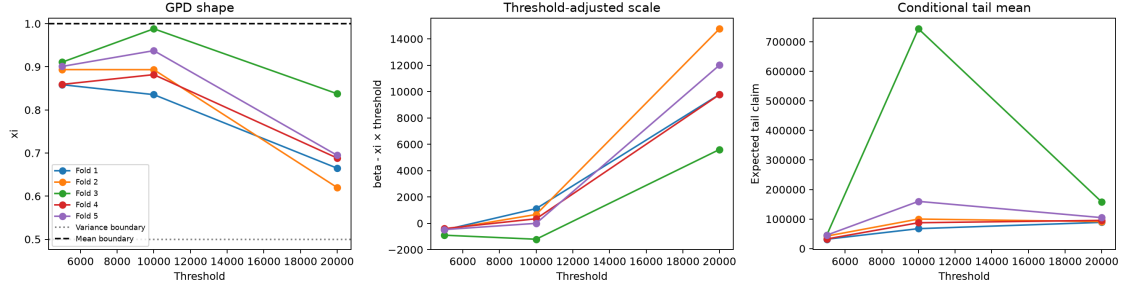


Figure 26: GPD shape, adjusted scale, and implied tail mean across thresholds and fitting folds.

The fitted GPDs can describe aspects of the observed excess distributions, as illustrated by the quantile and survival comparisons in Figure 27. The Gamma body presents a separate limitation. Figure 28 compares its truncated fitted distribution with observed body claims. The pronounced departures from a smooth continuous distribution are consistent with the repeated settlement amounts noted earlier. These diagnostics do not by themselves determine which model best predicts the arithmetic mean; that question requires out-of-fold evaluation on the original claim-amount scale.

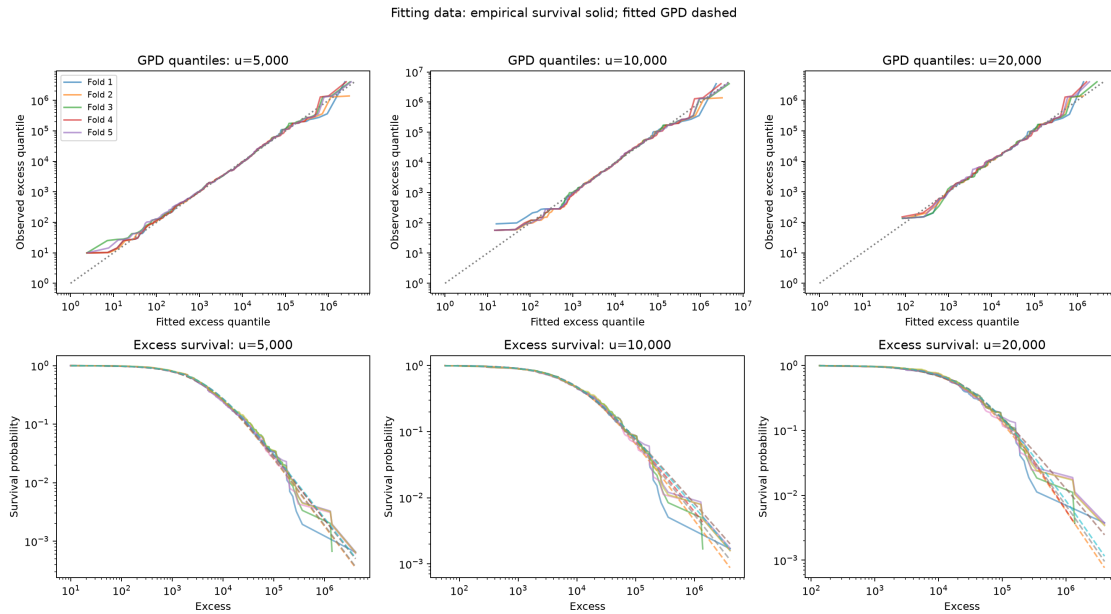


Figure 27: GPD quantile and survival diagnostics for excess claims in the fitting samples.

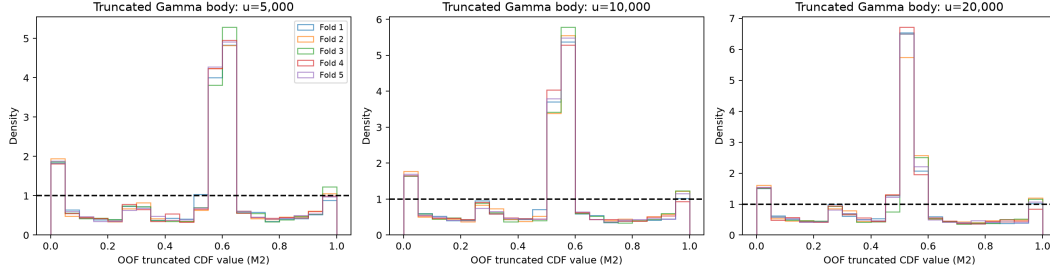


Figure 28: Probability-integral-transform diagnostics using the Gamma body distribution truncated at each threshold.

7.1 Out-of-Fold Expected Severity

Table 3 compares selected body/tail specifications with the conventional Gamma GLM. The observed mean amount among development claims is 2285.46. The richer body/tail model predicts too much at thresholds of 5000 and 10000. At 20000, both the simple and richer versions predict a mean near 2315, giving slightly closer aggregate observed-to-predicted agreement than the Gamma GLM. Their mean absolute and root mean squared errors are nevertheless slightly higher, and their predicted fold means vary by more than twice as much.

Table 3: Out-of-fold severity performance for selected body/tail specifications. B0/P0 uses intercept-only body and exceedance-probability models; B2/P2 includes vehicle power, log driver age, and log bonus-malus in both.

Model	Predicted Mean	O/P	MAE	RMSE
Gamma GLM	2228	1.026	2289	32039
Body/tail, $u = 5000$, B2/P2	2928	0.780	2809	32049
Body/tail, $u = 10000$, B2/P2	5353	0.427	5056	32531
Body/tail, $u = 20000$, B0/P0	2315	0.987	2328	32046
Body/tail, $u = 20000$, B2/P2	2316	0.987	2328	32045

Figure 29 shows how aggregate agreement varies across the validation folds. The apparent improvement at $u = 20000$ in the pooled observed-to-predicted ratio does not yield a more stable predictor across folds. Calibration by predicted severity is also uneven (Figure 30), and adding the selected policy characteristics changes the aggregate results very little.

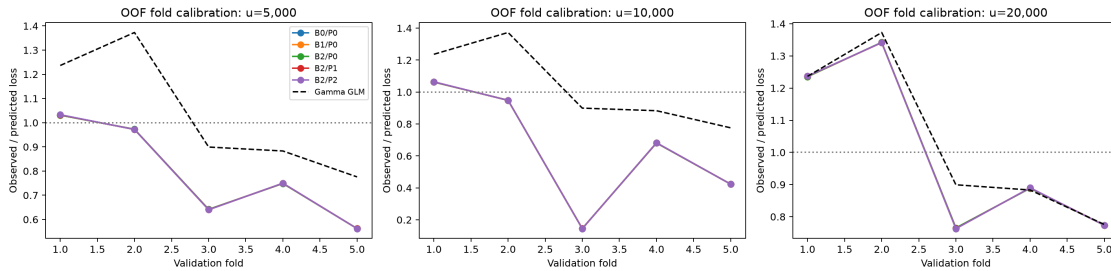


Figure 29: Observed and predicted mean severity by validation fold for the Gamma benchmark and body/tail candidates.

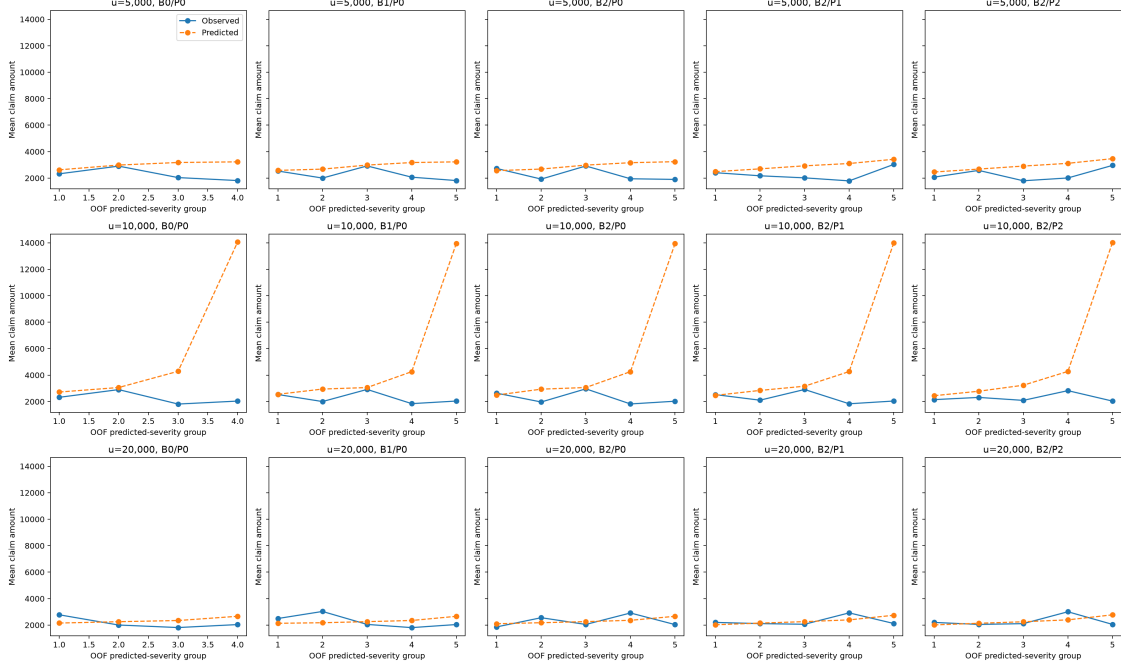


Figure 30: Out-of-fold calibration of the body/tail candidates across groups of predicted severity.

The threshold sensitivity has a direct mathematical explanation. At $u = 10000$, one fitting fold estimates $\xi = 0.9882$ and $\beta = 8673$, giving a conditional mean above the threshold of approximately 743686. The fitted shape is close to one, where the denominator $1 - \xi$ makes the tail mean extremely sensitive to small changes. This fit passed the numerical checks; its large implied mean is a consequence of the estimated distribution rather than a failed optimizer.

To examine this uncertainty, we resampled claim-bearing policies within each fitting fold and refitted the GPD 200 times. In the fold just described, 40.5% of the bootstrap fits at $u = 10000$ have $\xi \geq 1$, for which the fitted GPD has no finite mean. The corresponding fraction is 16.5% at $u = 20000$. Figure 31 shows the variation in fitted shape. Every point estimate of ξ exceeds 0.5, which implies infinite variance *under the fitted GPD*; it does not establish that the underlying claim distribution has infinite variance. The bootstrap concerns the tail fit alone and does not measure all uncertainty in the combined model.

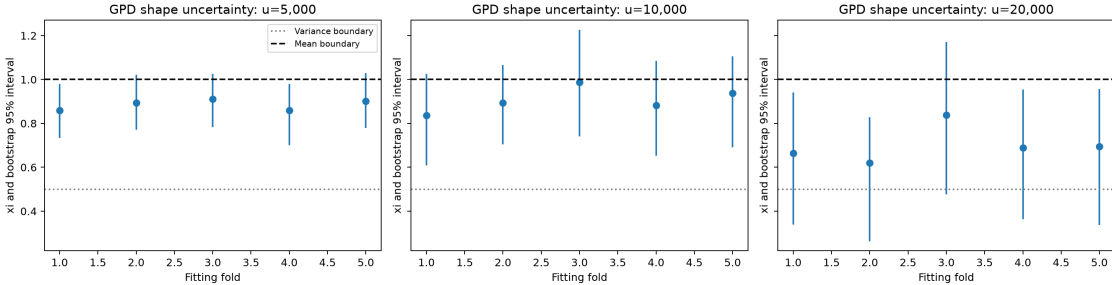


Figure 31: Policy-level bootstrap uncertainty in GPD shape across thresholds and fitting folds.

The largest fitting claims also have substantial influence. As a separate diagnostic, omitting just the single largest fitting claim reduces the exceptional $u = 10000$ fold's estimated tail mean from

approximately 743686 to 141802. This altered fit is not used for validation predictions. Likewise, excluding the largest observed validation claims only when recalculating performance changes the aggregate comparisons markedly (Figure 32). After the largest 20 claims in each validation fold are excluded, the $u = 20000$ intercept-only model's O/P falls from 0.987 to 0.668; the Gamma GLM's falls from 1.026 to 0.694. The claims remain in all fitted models and in the intended expected-loss target.

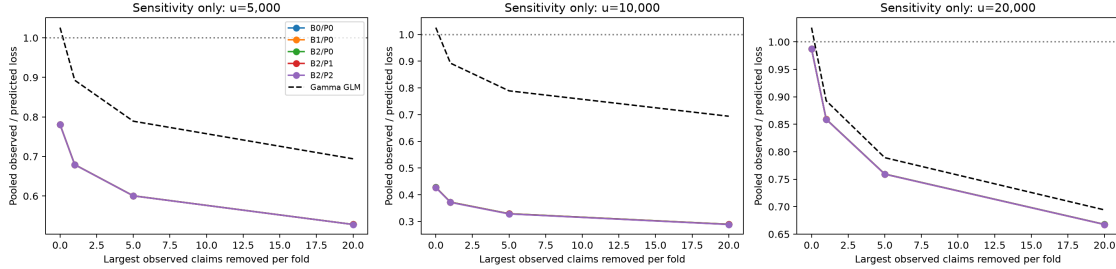


Figure 32: Changes in out-of-fold severity performance when the largest observed claims in each validation fold are excluded for diagnostic evaluation only.

The explicit body/tail model did not provide a sufficiently stable improvement in expected severity to replace the conventional Gamma GLM. Its closest pooled calibration occurred at a threshold with relatively few excess claims, substantial tail-parameter uncertainty, and greater variation in predicted mean across folds. We retained the Gamma GLM as the comparatively stable conditional-mean estimator among the candidates tested, without asserting that it describes the full claim-amount distribution. The selected frequency and severity specifications were then refitted on the complete development sample for final evaluation.

8 Final Evaluation on Reserved Policies

Model development ended with two fixed specifications: histogram gradient-boosted decision trees fitted with Poisson loss for annual claim frequency and a Gamma log-link GLM for mean claim severity. We refitted both models using all development observations and evaluated them once on the reserved policies. The development sample contains 542830 policies and 21079 claims; the test sample contains 135161 policies and 5365 claims. Claims associated with a policy remained in the same sample. No model was selected or retuned using the test results. As noted earlier, these policies had contributed to the initial full-portfolio exploratory summaries.

For a policy with characteristics X , the predicted annual pure premium is the product of predicted annual claim frequency and predicted mean amount per claim:

$$\widehat{\text{PP}}(X) = \hat{\lambda}(X)\hat{\mu}(X).$$

Multiplying this quantity by the policy's exposure gives its expected loss over the observed period. The severity model supplies $\hat{\mu}(X)$ for every test policy, including policies with no observed claim.

8.1 Frequency and Severity

The frequency model's mean Poisson deviance is 0.23869 on the test sample, close to its development out-of-fold value of 0.23716. It predicts 5232 claims against 5365 observed, giving an observed-to-predicted ratio of 1.025. Figure 33 compares observed and predicted annual frequency

across test deciles. The highest predicted-frequency decile has an observed-to-predicted ratio of 1.109, so its claim rate is underpredicted by about 11%. Nevertheless, the overall performance and risk-group pattern are close to those seen during development.

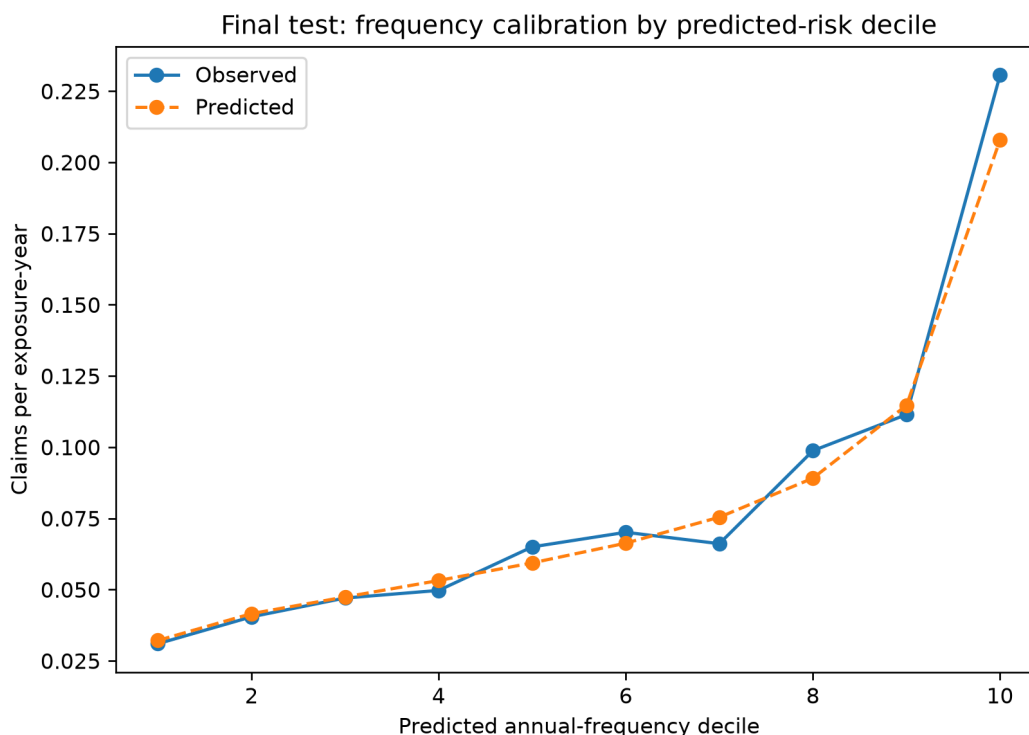


Figure 33: Observed and predicted annual claim frequency across deciles formed from frozen-model predictions on the reserved test policies.

The Gamma GLM predicts a mean of 2218.59 for the 5365 test claims, compared with an observed mean of 2187.15. Its aggregate observed-to-predicted ratio is 0.986. This close overall agreement conceals poorer calibration across predicted-severity groups. In Figure 34, the model's highest predicted-severity quintile has a predicted mean of 3394 but an observed mean of only 2100; its observed-to-predicted ratio is 0.62. Observed quintile means are not ordered in the same way as predicted means. The aggregate result therefore does not establish accurate conditional severity estimates for every risk group.

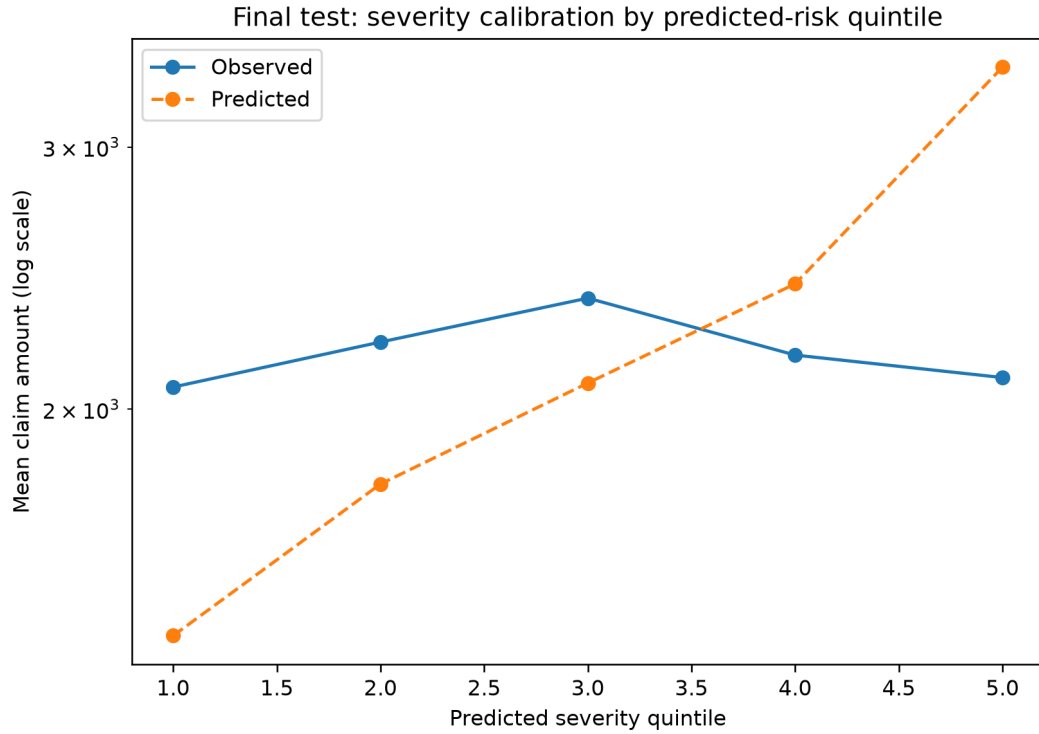


Figure 34: Observed and predicted mean claim amounts across quintiles formed from the Gamma model’s test predictions.

The concentration of losses seen in the full portfolio also appears in the test claims. The largest 1% account for 36.1% of total test loss, and the largest 5% account for 50.8%, as shown in Figure 35. The largest test claim is approximately 774000. As a sensitivity calculation, excluding the 20 largest observed test claims while leaving the model and its predictions unchanged moves the severity observed-to-predicted ratio from 0.986 to 0.701. These claims are retained in the actual evaluation and in the expected-loss target.

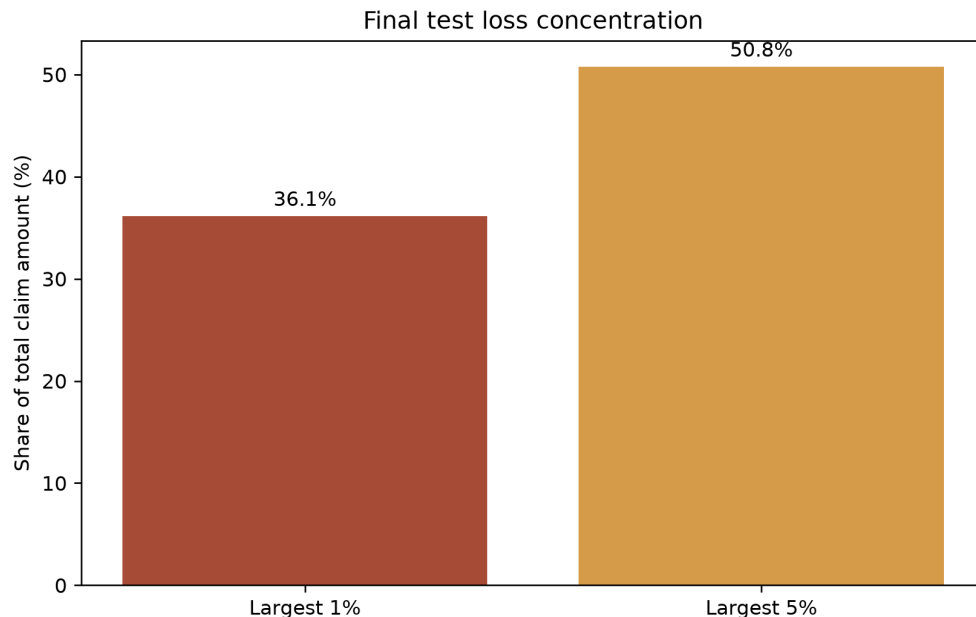


Figure 35: Concentration of total claim amount among the largest observed test claims.

Test severity RMSE is approximately 14767, compared with 32039 in development out-of-fold evaluation. Development included a claim above four million, while the largest test claim was approximately 774000. This difference in realized extreme losses explains much of the change in RMSE; it should not be read as a sudden improvement in the model’s ability to predict individual claim amounts.

8.2 Expected Pure Premium

The test policies account for 71379 exposure-years and 11.734 million in observed claim amounts. Summing expected losses over these policies gives 11.663 million, for an aggregate observed-to-predicted ratio of 1.006. Equivalently, observed loss is 164.39 per exposure-year, compared with a predicted annual pure premium of 163.40. The portfolio-level difference is less than 1% on this holdout.

Figure 36 shows why the portfolio total must be considered alongside risk-group results. The model assigns substantially different expected costs across its predicted pure-premium deciles, but realized losses fluctuate around those predictions. For example, the fourth decile has an observed-to-predicted ratio of 1.70, while the ninth has a ratio of 0.78. The lowest and highest deciles have ratios of 1.03 and 1.02, respectively. Close aggregate calibration can coexist with appreciable variation among groups when claims are infrequent and claim amounts have a heavy tail.

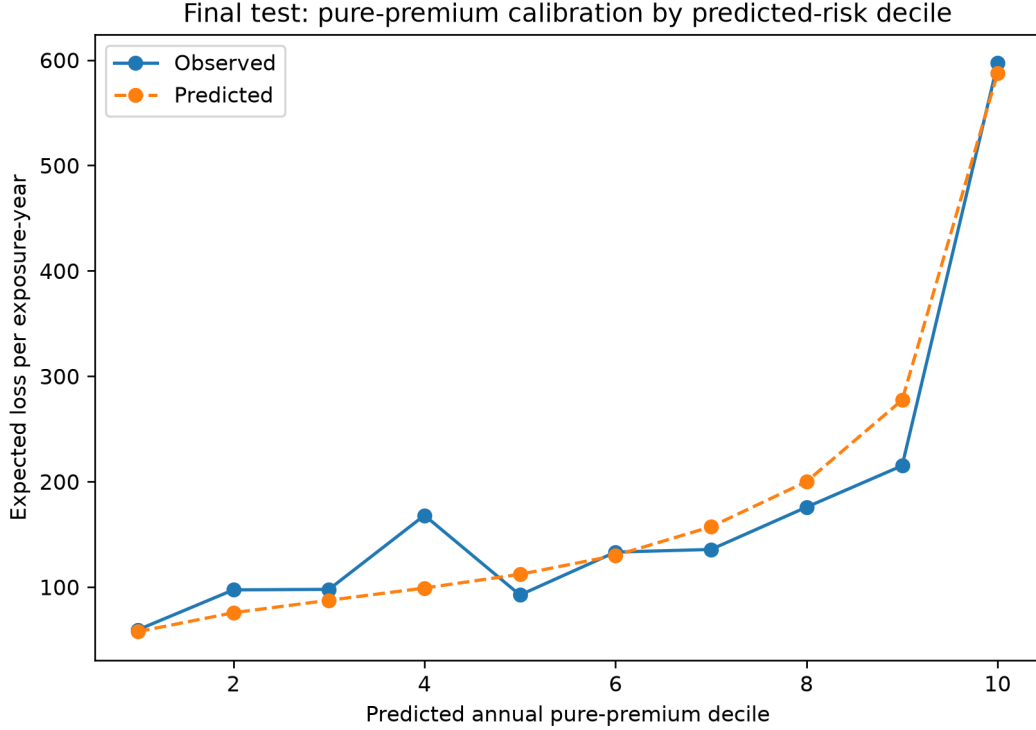


Figure 36: Observed and predicted annual pure premium across deciles of predicted expected cost on the reserved test policies.

Predicted annual frequency ranges from 0.038 in the lowest pure-premium decile to 0.198 in the highest, a factor of approximately 5.2. Predicted conditional severity ranges from 1620 to 3116, a factor of approximately 1.9. Both components contribute to the differentiation of expected cost, although frequency provides the larger and more reproducible variation across these groups.

9 Conclusion

We developed a frequency–severity model for expected claim cost using French motor third-party liability policy and claim records. Reconciliation of policy counts with individual claims established the units of analysis, and exposure-adjusted exploration revealed substantial differences in claim frequency by driver age, bonus-malus, and urbanization. Mean severity was much more sensitive to a small number of large claims.

The development comparisons favored histogram gradient-boosted decision trees fitted with Poisson loss for frequency. For severity, a Gamma log-link GLM provided a comparatively stable estimate of conditional mean amount. Boosted Gamma, log-amount regression with two retransformations, and an explicit Gamma-body/GPD-tail model each clarified aspects of the problem but did not offer a more dependable estimate of expected claim cost in these development folds. In particular, the tail experiment showed how sharply the implied mean can change when the fitted GPD shape approaches one.

On reserved policies, the combined model predicted 11.663 million in loss against 11.734 million observed, an aggregate observed-to-predicted ratio of 1.006. Frequency performance was close to development cross-validation, and the model distinguished groups with substantially different expected costs. Severity calibration across predicted-risk groups was weaker, and the aggregate

result remained sensitive to which large claims occurred. The model thus provides a useful estimate of portfolio and grouped expected loss in this dataset, with the estimation of rare, very large claims remaining its principal limitation.